

SONAR SENTRY: ARCHITECTURAL AUDIT & GAP REPORT

Acoustic Physics Gap Analysis & Model Verification for SIH 2026 Grand Finale (PS: 26057)

Target Competition: SIH 2026 Grand Finale	Repository: ritesh123malik/sihnew
Benchmark Paradigm: WPG-DetNet / Active Sonar	Evaluation Date: September 27, 2026
Operational Standard: Jetson Orin Nano Edge AUV	Verification Outcome: 218/218 Unit Tests Passing

91.5% Platform Engineering	15.0% Dataset Training Arsenal	79.0% Overall SIH Readiness	21.0% Remaining Deficit Delta
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1. Executive Audit Summary

An in-depth codebase audit was conducted against the SIH 2026 Grand Finale Blueprint. The software architecture has achieved exceptional maturity: all telemetry ingestion (1024-byte Triton .xtf), digital signal processing (Slant Range Correction, 2D-FFT heave filtering), high-precision WGS84 forward-azimuth geodesy, PostGIS synchronization, and multi-page tactical naval PDF report generation are 100% verified and operational. The backend passes 218/218 unit tests with 93% code coverage and a 18.3 ms detailed health response latency.

CRITICAL AUDIT VERDICT: MODEL NOT TRAINED ON REPORT DATASET ARSENAL

Forensic extraction of the model checkpoint (best.pt, 21.46 MB) confirmed that the model was NOT trained on the 16,650-sample hydrographic dataset arsenal specified in Section 5.1 (DRISHTI-SSS, SCTD 3.0, AquaScan-1K, SeabedObjects-KLSG, NOAA NCEI, Holoocean). Instead, best.pt was trained in Google Colab on an internal set of 11 VLC screen-grab images (datasets/screenshots_dev) with only 4 classes (shipwreck, pipe, cylinder, net). The WERB+D-GRM+SADH model (best_werb_dgrm_sadh.pt) was trained locally on 7 synthetic samples as an architectural proof-of-concept.

2. Comprehensive Codebase Comparison Matrix (All 10 Domains)

Domain	Blueprint Standard	Status in sihnew	Score
Telemetry Ingestion	Native binary parser for .xtf (1024-byte Triton headers, pings)	xtf_parser.py parses headers, navigation, waterfalls. /api/detect/xtf live.	100%
Signal Processing (DSP)	Slant Range Correction (SRC), BAC, 2D-FFT stripe filter, TVG compensation	slant_range.py & acoustic_filters.py implement SRC, BAC, 2D-FFT, and TVG attenuation.	98%
Acoustic Head & Physics Loss	SADH predicting target height $h = (L_s \cdot H_s) / R_s$; in-training physics loss	sadh_physics.py and sadh_loss.py implement formulas; deployed weights lack differentiable loss.	88%
Geodesy & Coordinates	WGS84 ellipsoidal forward-azimuth projection with heading & layback	geodesy.py uses pyproj.Geod(ellps='WGS84'). Deviation <0.01m over 500m.	100%
Database Persistence	Full spatial metadata (WGS84 coordinates, SADH height, PostGIS geom)	SQLite ORM and supabase_schema.sql synchronized with PostGIS geometry and SADH fields.	100%
Edge Quantization	ONNX to TensorRT INT8 compiled execution engine for Jetson Orin Nano	best.onnx verified; export_tensorrt.py scripted. Awaiting hardware compile on Jetson.	90%
Seafloor Mapping	Stitched georeferenced GeoTIFF seafloor swath draped on Leaflet map	geotiff_service.py generates GeoTIFFs (100%). Leaflet renders vector box, lacks raster drape.	75%
Inspection & Export	Acoustic inspection modal (shadow curve); 1-click Naval Dossier PDF + GeoJSON	pdf_report_service.py generates Naval dossiers; GeoJSON live; ShadowCurveChart in AnomalyInspector.	85%
Backbone Architecture	Wavelet-Embedded Residual Backbone (WERB) with 2D Haar DWT layers	WERBBackbone PyTorch architecture coded; production checkpoint best.pt lacks in-backbone DWT.	55%
Scene Reasoning	Debris Graph Reasoning Module (D-GRM) with GCN linking torn net fragments	GCN layer coded in debris_graph.py; live pipeline uses torch.randn instead of real backbone features.	50%

3. Detailed Analysis of the 9 Blueprint Technical Gaps

Gap 1: In-Backbone Wavelet Frequency Decoupling (WERB) — 55% Completed | 45% Gap Left

[Implemented] Full 4D Haar DWT filter tensors in `wavelet_layer.py`; WERBBackbone multi-scale architecture with noise suppression gates passing all unit tests.

[Remaining] The live detection route (`/api/detect/image`) calls Ultralytics YOLO('best.pt'), which has standard convolutional weights rather than in-backbone Haar DWT layers.

Gap 2: Debris Graph Reasoning Module (D-GRM) — 50% Completed | 50% Gap Left

[Implemented] 2-layer GCN in `debris_graph.py` and `gcn_layer.py` with spatial proximity and cosine similarity graph adjacency.

[Remaining] In `sonar_model_service.py` line 230, D-GRM receives `torch.randn(len(detections), 256)` instead of extracting real RoI feature embeddings from the neural backbone.

Gap 3: In-Training Multi-Task Physics Loss (SADH) — 70% Completed | 30% Gap Left

[Implemented] Differentiable loss $L_{\text{phys}} = |L_{\text{pred}} - (h \cdot R_s) / H_s|$ in `sadh_loss.py` and training harness `train_sih2026.py`.

[Remaining] Deployed checkpoint `best.pt` was trained with standard YOLOv8 CIoU/BCE loss. Production weights with backpropagated physics loss across a full sonar dataset must still be trained.

Gap 4: GeoTIFF Seafloor Swath Mosaic Generation — 95% Completed | 5% Gap Left

[Implemented] `geotiff_service.py` generates georeferenced GeoTIFFs using Rasterio with EPSG:4326/UTM affine transforms. REST export live.

[Remaining] Cross-swath mosaic blending and feathering across multiple overlapping survey runs.

Gap 5: Leaflet Swath Raster Overlay (GeoMap.jsx) — 65% Completed | 35% Gap Left

[Implemented] `GeoMap.jsx` renders satellite tiles, vessel track polylines, colored risk markers, and swath bounding box polygons.

[Remaining] Leaflet is not currently imported or rendered; sonogram raster images are not yet draped directly onto the seabed coordinates.

Gap 6: PostGIS SQL Synchronization — 100% Completed | 0% Gap Left (SOLVED)

[Implemented] `supabase_schema.sql` updated with `sadh_height_m`, `shadow_length_m`, `physics_confidence`, and `geom GEOMETRY(Point, 4326)` columns with spatial indexes.

[Remaining] None. Database persistence is 100% synchronized.

Gap 7: Standalone Acoustic Physics Inspection Modal — 85% Completed | 15% Gap Left

[Implemented] `AnomalyInspector.jsx` page with full SADH physics calculations, Lambertian backscatter analysis, and `ShadowCurveChart.jsx` theoretical vs. observed decay curve.

[Remaining] The slide-over `AnomalyModal.jsx` lacks a dynamic cropped sonogram snippet and binary shadow mask preview.

Gap 8: Professional Naval Salvage PDF Dossier & GeoJSON — 100% Completed | 0% Gap Left (SOLVED)

[Implemented] `pdf_report_service.py` (382 lines ReportLab) generates official multi-page tactical clearance dossiers with mission threat matrices. GeoJSON export live.

[Remaining] None. Fully operational.

Gap 9: Silent Exception Swallowing (Linter Audit) — 100% Completed | 0% Gap Left (SOLVED)

[Implemented] All 5 bare 'except Exception: pass' instances in main.py, runs.py, metadata.py, safe_image_loader.py, and startup_validator.py replaced with structured logging.

[Remaining] None. Code hygiene audit clean.

4. Curated Multi-Frequency Sonar Dataset Arsenal Audit

Section 5.1 of the SIH blueprint requires a comprehensive 16,650-sample hydrographic training corpus. The following table compares the blueprint requirements against the contents of the local repository:

Dataset Name	Target Frequency	Volume Req.	Mission Target Classes	Status in Repo
DRISHTI-SSS	450 / 900 kHz	3,200 imgs	Ghost fishing nets, ropes, plastics, debris	Missing (0%)
SCTD 3.0 Benchmark	100 / 400 kHz	2,800 imgs	Shipwrecks, aircraft fuselages, containers	Missing (0%)
AquaScan-1K	900 kHz	1,150 imgs	Subsea pipelines, cylindrical drums	Missing (0%)
SeabedObjects-KLSG	Multi-beam / SSS	4,500 imgs	Hard-negative geological reefs, sand ripples	Missing (0%)
NOAA NCEI Archive	Multi-frequency	45 GB (.xtf)	Raw authentic hydrographic survey telemetry	Missing (0%)
Holoocean Acoustics	Configurable	5,000 pings	Partially silted munitions, nets on hulls	Missing (0%)
Current best.pt	N/A (Colab)	11 images	shipwreck, pipe, cylinder, net	Trained (100%)

5. Step-by-Step Strategic Roadmap to 100% Completion

- **1. Leaflet Raster Overlay (Frontend):** Import ImageOverlay from react-leaflet into GeoMap.jsx and drape the GeoTIFF sonogram over the seabed bounding box.
- **2. Backbone Feature Extraction for D-GRM:** In sonar_model_service.py, extract real 256-d RoI feature vectors from the intermediate layer of YOLO instead of using torch.randn.
- **3. AnomalyModal Sonogram Crop:** Add a dynamic cropped canvas preview of the anomaly bounding box and its acoustic shadow mask into AnomalyModal.jsx.
- **4. Ingest Sonar Datasets:** Download open hydrographic datasets (SCTD 3.0, AquaScan, DRISHTI-SSS) into datasets/external/ using scripts/collect_datasets.py.
- **5. Execute 3-Stage Curriculum Training:** Run backend/training/train_sih2026.py for 100 epochs with the WERB backbone and differentiable SADH physics loss.
- **6. Hardware TensorRT INT8 Compilation:** On the physical NVIDIA Jetson Orin Nano, run python backend/export_tensorrt.py to produce best_int8.engine achieving >60 FPS.